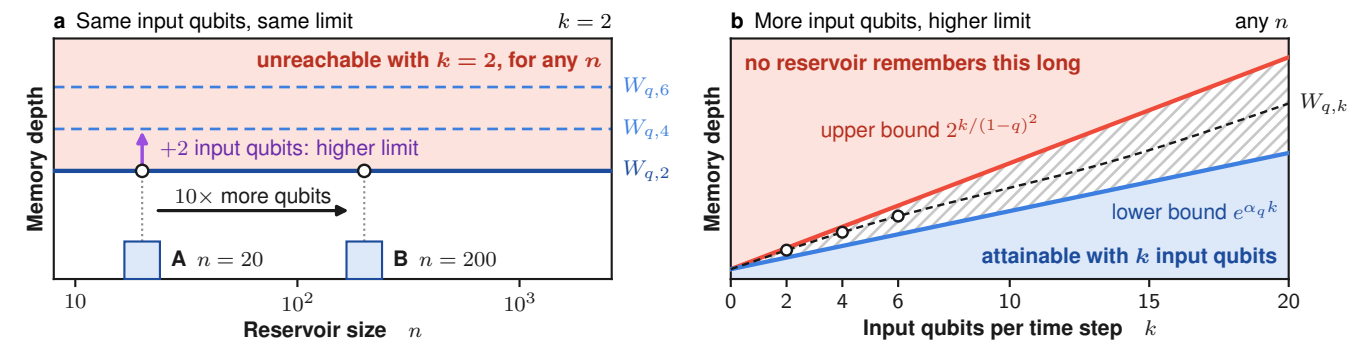

a Same input qubits, same limit
$k = 2$
unreachable with $k = 2$, for any n
$W_{q,6}$
$W_{q,4}$
$W_{q,2}$
+2 input qubits: higher limit
10× more qubits
A $n = 20$
B $n = 200$
Memory depth
10
10^2
10^3
Reservoir size n
b More input qubits, higher limit
any n
no reservoir remembers this long
upper bound $2^{k/(1-q)^2}$
$W_{q,k}$
lower bound $e^{\alpha_q k}$
attainable with k input qubits
Memory depth
0
2
4
6
10
15
20
Input qubits per time step k